Diseases of carnation

1. Damping off - *Pythium* spp, other fungi
2. Wilt - *Fusarium oxysporum* f.sp. *dianthi*
3. Canker, stub die-back, - *Fusarium avenaceum,*
   stem or crown-or root-rot - *F. culmoram, F. roseum*
4. Bacterial wilt - *Pseudomonas caryophylli*

Diseases of anthurium

1. Anthracnose - *Colletotrichum gloeosporioides*
   spadix rot / black nose PS*: Glomerella cingulata*
2. Root rot - *Pythium splendens,* other fungi
3. Powdery mildew - *Erysiphe communis*

Carnation damping off - *Pythium* spp

Carnation damping off - *Pythium* spp

Symptoms

- Caused by a **complex of fungi** that spreads on the surface of the soil, **damping off attacks seedlings** shortly after they emerge and **form their first pair of leaves** .
- The stem is damaged at soil level, causing **seedling to collapse and rot.**
- **In 24hrs a whole tray of seedlings** can be destroyed.
- Overcrowding encourages damping off, as does a high level of humidity.

Carnation damping off - *Pythium* spp

Management

- Use a fresh batch of **sterilized media for raising seedlings** for each planting and **avoid over watering**.
- **Drenching the seed mix** with a solution of copper oxychloride before and after sowing.
- **New cuttings** may also be affected by damping off.

Carnation wilt - *Fusarium oxysporum* f.sp.*dianthi*

Wilted, yellowing plant in center of photograph has fusarium

Carnation wilt - *Fusarium oxysporum* f.sp.*dianthi*

Carnation wilt - *Fusarium oxysporum* f.sp.*dianthi*

Fusarial wilt **widely prevalent** in carnation all over the world

Symptoms

- First noticed during **mid day in the polyhouses** when the temp is high, which are not visible when the temp goes down
- The initial symptoms are **foliar yellowing** and **production of 'crook neck' or bent shoots** (Top of main shoot grows at right angle to main stem)
- First symptoms are a slow wilting of shoots, **often on only one side of the plant**
- Wilting is accompanied by a discoloration of leaves, at first a **light gray-green and finally a pale yellow color**.

Carnation wilt - *Fusarium oxysporum* f.sp.*dianthi*

Symptoms contd…

- Initially the symptoms are confined to **part of the plant but ultimately on** the entire plant
- Stems when cut open show **brown discolouration** at the vascular region.
- Sometimes **pith and cortex** also get discoloured.
- Light **brown vascular discoloration** was observed at the base of the of the stem.
- When older plants are infected, similar symptoms are produced but the older leaves may **show chlorosis** followed by an **indistinct purple-red discoloration**.

Carnation wilt - *Fusarium oxysporum* f.sp.*dianthi*

Management: - Physical methods

- Soil **solarisation for nearly seven weeks** during summer months.
- Subsequent application of N fertilizers increases disease levels again, there by indicating that **N deficiency delays disease development** but does not provide a lasting protection against wilt pathogen.
- ***Trichoderma aureoviride*** is most effective against *F.oxysporum* f.sp.*dianthi*.

Carnation wilt - *Fusarium oxysporum* f.sp.*dianthi*

Management cond…

- Use of ***Pseudomonas putida*** have shown potential against *F.oxysporum* f.sp.*dianthi.*
- Nutrient management.
- Soil drenching by carbendazim.
- Application of **strobilurins** (Azoxystrobin, trifloxystrobin) showed high efficiency when applied as soil drenching.

Carnation - Canker, die-back, stem or crown-or root-rot - *Fusarium roseum* (PS: *Gibberella zeae*), *F. culmoram*

Fusarium root and crown rot on this carnation.

Crown rot of this *Dianthus* sp.

Carnation - Canker, die-back, stem or crown-or root-rot - *Fusarium roseum* (PS: *Gibberella zeae*), *F. culmoram*

Symptoms:

- Fungus often **invades the stumps** left after the shoots are removed and cause die back
- **Rotting of stems near the base/ the branch stems**
- The infection is noticed on all parts of the plant.
- The disease occurs **due to injuries caused by continuous harvesting.**
- Most infection occurs on the **upper part of second year plants indicating that the fungus is air-borne**
- The **rotting of roots and stem causes a progressive** wilting and collapse of the plant.

Carnation - Canker, die-back, stem or crown-or root-rot - *Fusarium roseum* (PS: *Gibberella zeae*), *F. culmoram*

Symptoms contd…

- The diseased tissues turn **brown, red or crimson.**
- The branches show **die-back because of the infection through flowers are cut or plants are pinched** .
- It grows down roots, the side break of main stem, causing wilting and death of branches **without discolouration of inner vascular tissues.**
- The fungus cause rotting of stems near the base and sometimes the branch

Carnation - Canker, die-back, stem or crown-or root-rot - *Fusarium roseum* (PS: *Gibberella zeae*), *F. culmoram*

Management

- Spraying captan @ 2g/lt before after harvesting of flowers

Carnation Bacterial wilt - *Pseudomonas caryophylli*

Carnation Bacterial wilt - *Pseudomonas caryophylli*

- Causes wilt at high temperatures

Symptoms

- **Sudden wilting**, production of **greenish yellow foliage**
- **Initial leaf and stem discolouration** is usually followed by plant wilting and death.
- **Stem bases may also show internodal stem cracking**, developing into small areas of dead tissues.
- Cutting of diseased stem often reveals **brownish yellowish bacterial ooze**

Carnation Bacterial wilt - *Pseudomonas caryophylli*

Management

- This disease cannot be directly controlled by a chemical means, simply because no chemical products are available that can control the disease .
- However sanitary measures can prevent further infection.
- At an early stage of disease, the KPV-Metodon, biochemical testing method can be used **to detect infections in cuttings**.

  Two wild sps Dianthus *capitus* subsp. *andrzejowskianus* and *D.henteri* are resistant.

Diseases of anthurium

1. Anthracnose - *Colletotrichum gloeosporioides*
   spadix rot / black nose PS*: Glomerella cingulata*

2. Root rot - *Pythium splendens*

3. Powdery mildew - *Erysiphe communis*

Anthracnose/ spadix rot / black nose -*Colletotrichum gloeosporioides*, PS: *Glomerella cingulata*

Anthracnose/ spadix rot / black nose -*Colletotrichum gloeosporioides*, PS: *Glomerella cingulata*

a. Black nose disease on *Anthurium* causes spadix darkening.

Anthracnose/ spadix rot / black nose -*Colletotrichum gloeosporioides*, PS: *Glomerella cingulata*

- **Problematic in high rainfall areas and most damaging** disease

Symptoms

- Circular to irregular lesions on leaves
- The infection occurs on young plants soon after planting
- Infects the **individual flowers on spadix as tiny brownish** spots
- Spadices and flowers are damaged
- During high humidity these spots enlarge, appear water-soaked and turn necrotic.

Anthracnose/ spadix rot / black nose -*Colletotrichum gloeosporioides*, PS: *Glomerella cingulata*

Symptoms contd…

- Sometimes the **entire spadix will turn black as lesions coalesce.**
- The shape of most lesions is, however, **angular due to the shape of the spadix tissue, eventually spadix fall off**
- As the disease becomes more severe, masses of orange coloured spores form on necrotic areas.
- Leaves and spathes are rarely if ever infected.
- Warm wet weather favours the outbreak of the disease which spreads fast in densely planted beds.

Anthracnose/ spadix rot / black nose -*Colletotrichum gloeosporioides*, PS: *Glomerella cingulata*

Management

- Sanitation
- Use of common fungicides (0.1% carbendazim + 0.2% copper oxychloride or 0.1% thiophanate methyl or 0.2% mancozeb or 0.15% kitazin) at fortnightly intervals helps in management of the disease.

Anthurium root rot - *Rhizoctonia similis*

Damping off of Anthurium - *Pythium splendens*

Figure 11. *Anthurium* wilt caused by *Phytophthora*.

Figure 12. *Pythium* and *Phytophthora* cause root sloughing.

Root rot - *Pythium splendens*

- *P. splendens*, *P. spinosum*, *P. vexans*, *Rhizoctonia* sp and *Calonectria crotalariae* were isolated from the plants showing root rot symptoms in Hawaii, USA
- Causes serious losses especially during rainy season

Symptoms

- **Stunted growth, reduced leaf size and flowers**
- Lack of **leaf and flower lusture**
- General **reduction in vigour**

Anthurium Powdery Mildew- *Erysiphe communis*

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Anthurium Powdery Mildew- *Erysiphe communis*

- This disease, caused by *Erysiphe communis* was first reported on ***Anthurium scherzerianum* grown in Germany.**
- The pathogen covers the upper leaf surface with white powdery growth.
- Powdery growth also appears on the stem.
- Spraying with fungicides dinocap (0.06%), carbendazim (0.05%) or hexaconazole (0.05%) or difenoconazole (0.03%) and repeat at 10-14 days interval